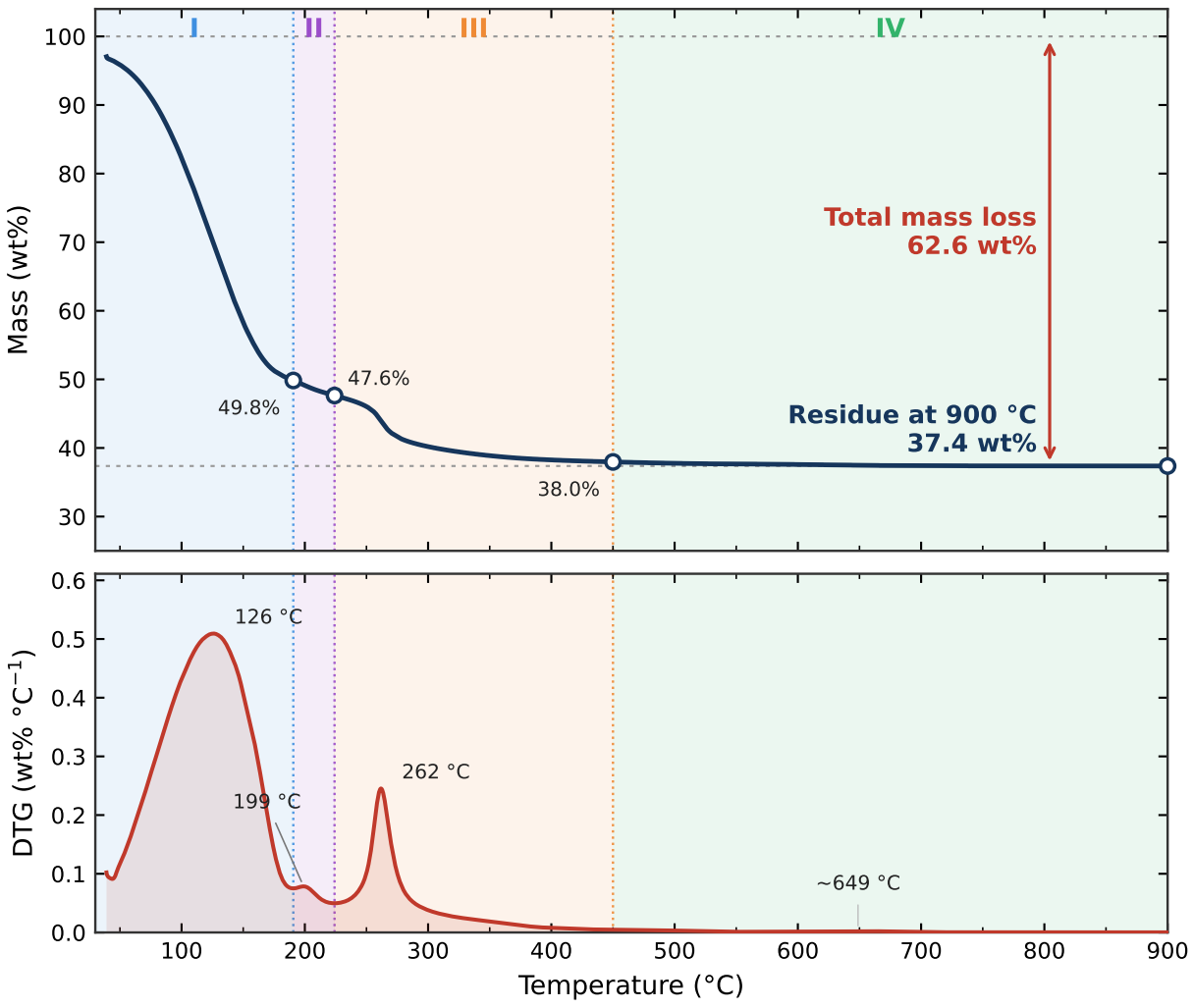
TGA / DTG of zeolite 13X paste – Batch 7 (C1)

Paste (wet basis): 13X 37.77 | bentonite 4.42 | PEG-400 7.73 | methyl cellulose 1.12 | water 48.96 wt% • Air, 50 mL/min, 10 °C/min



Air, 50 mL/min, 10 °C/min, 39→900 °C. Step limits at DTG minima (191 and 224 °C) and at 450 °C (DTG < 0.005 wt% °C⁻¹). Mass is relative to the sample loaded in the pan; 3.0 wt% water evaporated before the first data point.

Mass-loss steps

Step	T range	Loss (wt%)	Assignment
I	RT–191 °C	50.2	Free water of the paste + adsorbed water of 13X and bentonite [4, 5]
II	191–224 °C	2.2	Strongly bound zeolite water + onset of PEG-400 oxidation [1, 2, 4]
III	224–450 °C	9.7	Oxidative burn-off of PEG-400 and methyl cellulose [1, 2, 3, 6]
IV	450–900 °C	0.60	Dehydroxylation of bentonite (4.7–4.9 % of clay mass) [5]
Total		62.6	Residue 37.4 wt%: dehydrated 13X + dehydroxylated bentonite

Expected from the paste composition

Component	Added (wt%)	Measured (wt%)
Water	48.96	50.2 (step I)
PEG-400 + methyl cellulose	8.85	11.9 (steps II + III)
Bentonite OH (upper bound)	0.21	0.60 (step IV)
Solids (13X + bentonite, if dry)	42.19	37.4 (residue)

The residue is 4.8 wt% lower than the dry-solids value because the 13X and bentonite were weighed with their own moisture (≈11 % of the solids). This water leaves in steps I–III, so these steps exceed the added water and organics.

References

[1] Marcos et al., Nanomaterials 8 (2018) 16. doi:10.3390/nano8010016

[2] Han, Kim, Kwon, Polymer 38 (1997) 317–323. doi:10.1016/S0032-3861(97)88175-X

[3] Xiao et al., Polymers 13 (2021) 3291. doi:10.3390/polym13193291

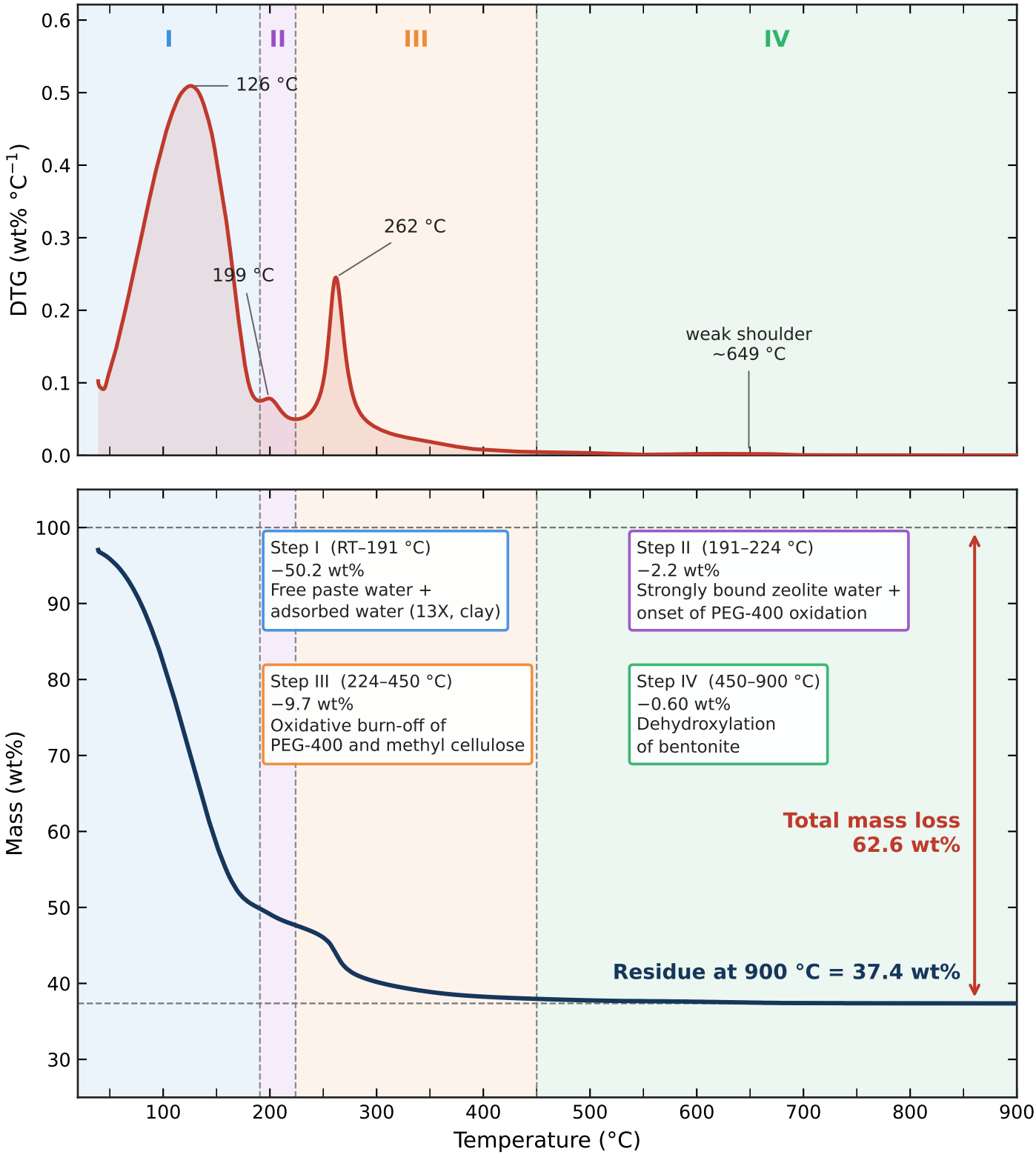
[4] de Gennaro et al., Materials 15 (2022) 5574. doi:10.3390/ma15165574

[5] Derkowski & Kuligiewicz, Clays Clay Miner. 70 (2022) 946–972. doi:10.1007/s42860-023-00222-y

[6] Guo et al., Coatings 10 (2020) 499. doi:10.3390/coatings10050499

TGA / DTG of zeolite 13X paste – Batch 7 (C1)

13X 37.77 | bentonite 4.42 | PEG-400 7.73 | methyl cellulose 1.12 | water 48.96 wt%



Air, 50 mL/min, 10 °C/min, 39→900 °C. Step limits at DTG minima (191 and 224 °C) and at 450 °C (DTG < 0.005 wt% °C⁻¹).
Mass is relative to the sample loaded in the pan; 3.0 wt% water evaporated before the first data point.

TGA / DTG Complete Analytical Report – zeolite 13X paste – Batch 7 (C1)

Sample ID: Zeolite Paste B7 • Operator: Shayan • Date: 2026-10-28 • Raw file: Zeolite Batch 7 C1 SG.xlsx

1. Experimental & Measuring Conditions

Measuring Parameter / Condition	Value
Sample	zeolite 13X paste – Batch 7 (C1)
Sample ID (vault)	Zeolite Paste B7
Operator	Shayan
Date	2026-10-28
Raw data file	Zeolite Batch 7 C1 SG.xlsx
Analysis	Thermogravimetric analysis (TGA) with derivative curve (DTG)
Atmosphere	Air (oxidative)
Gas flow rate	50 mL/min
Heating rate (set)	10 °C/min
Heating rate (fitted from data, 100–800 °C)	9.91 °C/min
Temperature range	39.1 → 900.0 °C
Run time	87.2 min
Number of data points	10465
Sample mass (mg)	Not given in the export – fill in
Mass basis	wt% relative to the mass loaded in the pan (100 %)
First data point	97.04 wt% at 39.1 °C, i.e. 2.96 wt% water evaporated before recording started
Step limits	DTG minima at 190.7, 224.1, 450.0 °C [min, min, tail]
Total mass loss (from 100 %)	62.63 wt%
Mass loss in the recorded part only	59.67 wt%
Residue at 900 °C	37.37 wt%

2. Paste Composition (Wet Basis)

No.	Component	Mass (g)	wt% (wet)	Preparation Note
1	Zeolite 13X	43.97 g	37.77 %	
2	Bentonite	5.14 g	4.42 %	
3	PEG-400	9.00 g	7.73 %	5 g before + 4 g after
4	Methyl cellulose	1.31 g	1.12 %	
5	Water	57.00 g	48.96 %	50 g before + 7 g after
Total		116.42 g	100.00 %	

3. Quantitative Mass-Loss Steps Summary

Step	T start (°C)	T end (°C)	DTG peak (°C)	Mass start (%)	Mass end (%)	Mass loss (%)
I	RT (loading)	190.7	126.0	100.0	49.84	50.16
II	190.7	224.1	198.9	49.84	47.65	2.19
III	224.1	450.0	261.7	47.65	37.97	9.68
IV	450.0	900.0	648.8	37.97	37.37	0.6
Total	RT (loading)	900.0	126.0	100.0	37.37	62.63

TGA / DTG Interpretation & Mass Balance – zeolite 13X paste – Batch 7 (C1)

Sample ID: Zeolite Paste B7 • Atmosphere: Air, 50 mL/min, 10 °C/min

4. Step-by-Step Thermal Interpretation

Step	T range	Peak °C	Loss %	Assignment	Detailed Mechanism & Comment	Refs
I	RT–191 °C	126	50.16	Free water of the paste + adsorbed water of 13X and bentonite	Largest step. Evaporation of the mixing water; physisorbed water of 13X and the interlayer water of bentonite (which ends below ~200 °C) leave in the same range.	[4], [5]
II	191–224 °C	199	2.19	Strongly bound zeolite water + onset of PEG-400 oxidation	Small DTG peak. 13X keeps releasing bound water up to ~300 °C; PEG-400 starts to lose mass near 200–230 °C in N2 and earlier in air because it oxidises.	[1], [2], [4]
III	224–450 °C	262	9.68	Oxidative burn-off of PEG-400 and methyl cellulose	Sharp DTG peak. In N2, PEG-400 loses mass between ~187 and ~397 °C and methyl cellulose has T10 ≈ 292 °C and decomposes at ~343 °C; in air both burn earlier, which explains the peak below 300 °C.	[1], [2], [3], [6]
IV	450–900 °C	649	0.60	Dehydroxylation of bentonite (4.7–4.9 % of clay mass)	Very small, broad loss with a weak DTG shoulder. Smectite dehydroxylation occurs between ~300 and 900 °C with 4.7–4.9 % mass loss of the clay.	[5]

5. Expected from Paste Composition vs. Measured Mass Balance

Component / Balance Item	Expected %	Measured %	Analytical Note
Water	48.96	50.16	Step I is larger than the added water because the 13X and bentonite moisture also leaves here.
PEG-400 + Methyl cellulose	8.85	11.88	Steps II + III also contain strongly bound zeolite water.
Bentonite OH (4.42 × 4.8 %, upper bound)	0.21	0.6	Step IV.
Total mass loss (if 13X and bentonite were dry)	57.81	62.63	
Residue (if 13X and bentonite were dry)	42.19	37.37	The measured residue is lower because the raw 13X and bentonite contained moisture when weighed.
Moisture in 13X + bentonite as weighed (% of solids)		11.4	Estimate from the residue. If the paste dried before loading, the true value is higher. A TGA of the raw 13X powder would confirm it.

6. Literature References

[1] Marcos M.A., Cabaleiro D., Guimarey M.J.G., Comuñas M.J.P., Fedele L., Fernández J., Lugo L. PEG 400-based phase change materials nano-enhanced with functionalized graphene nanoplatelets. Nanomaterials 8 (2018) 16. (doi:10.3390/nano8010016)
Used for: PEG-400 in N2 at 10 K/min: onset ~507 K (234 °C); main loss 460–670 K (187–397 °C)

[2] Han S., Kim C., Kwon D. Thermal/oxidative degradation and stabilization of polyethylene glycol. Polymer 38 (1997) 317–323. (doi:10.1016/S0032-3861(97)88175-X)
Used for: PEG undergoes thermal/oxidative degradation in air

[3] Xiao Q., Huang M., Zhou X., Dai M., Zhao Z., Zhou H. Effect of molecular weight and degree of substitution on the physical-chemical properties of methylcellulose-starch nanocrystal nanocomposite films. Polymers 13 (2021) 3291. (doi:10.3390/polym13193291)
Used for: Methyl cellulose films in N2 at 10 °C/min: water loss below 100 °C, then chain degradation; T10 = 292 °C

[4] de Gennaro B., Cappi A., de Gennaro M., Bianco N., Langella A., Cappelletti P., Marocco A., Aprea P., Pansini M. Use of zeolites in the capture and storage of thermal energy by water desorption–adsorption cycles. Materials 15 (2022) 5574. (doi:10.3390/ma15165574)
Used for: Zeolite 13X: most of the zeolitic water is lost by 300 °C

[5] Derkowski A., Kuligiewicz A. Thermal analysis and thermal reactions of smectites: a review of methodology, mechanisms, and kinetics. Clays and Clay Minerals 70 (2022) 946–972. (doi:10.1007/s42860-023-00222-y)
Used for: Smectite (bentonite): dehydration ends below 200 °C; dehydroxylation between ~300 and 900 °C with 4.7–4.9 % mass loss

[6] Guo P., Wang F., Duo T., Xiao Z., Xu A., Liu R., Jiang C. Facile fabrication of methylcellulose/PLA membrane with improved properties. Coatings 10 (2020) 499. (doi:10.3390/coatings10050499)
Used for: Methyl cellulose in N2 at 10 °C/min: decomposition temperature 343 °C